

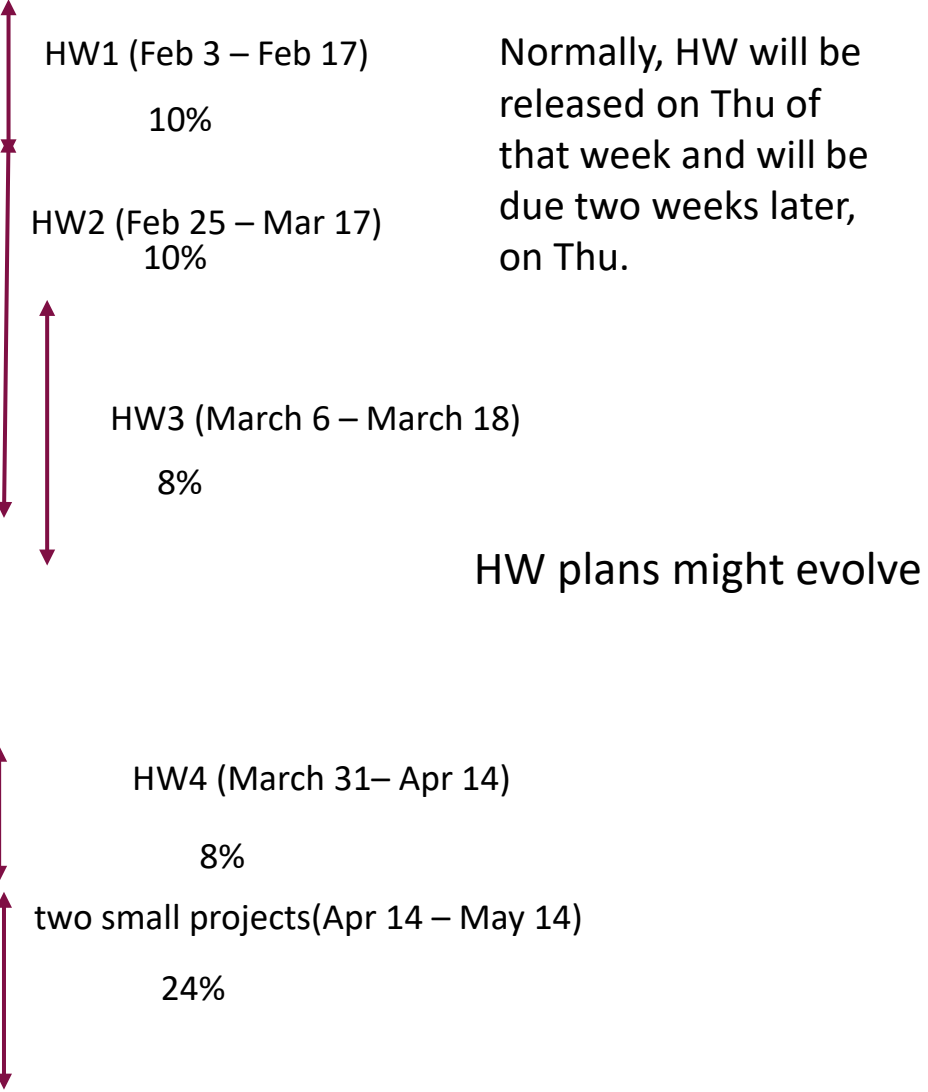
CIS 5800

Machine Perception

Instructor: Lingjie Liu

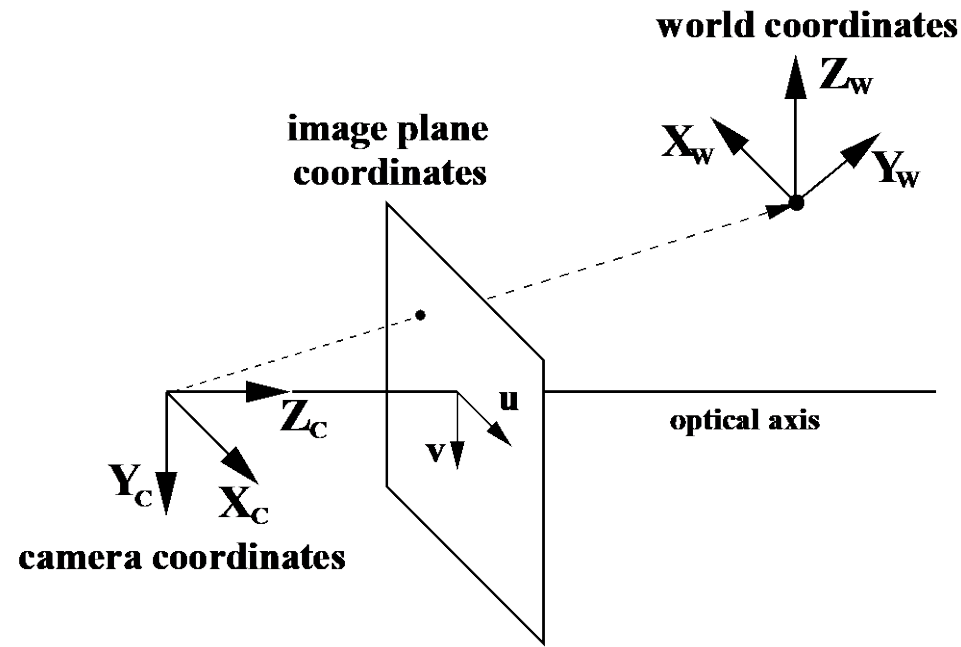
Lec 10 (Part 1): Feb 24, 2026

Week	Date*	Topics
1	Jan 20	
2	Jan 27	
3	Feb 3	
4	Feb 10	
5	Feb 17	
6	Feb 24	
7	Mar 3	
8	Mar 10 (Spring Break)	
9	Mar 17	
10	Mar 19	TA class (March 19); March 24, mid term exam review class. Midterm (20%, March 26);
11	Mar 31	
12	Apr 7	
13	Apr 14	
14	Apr 21	
15	Apr 28	
16	Btw May 4-12	Final Exam (20%)



Calibrating Camera Under More General Intrinsic

Recap: Perspective Projection Matrix



camera 3D coords to pixels Convert world to camera coordinates

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & & u_0 \\ & f & v_0 \\ & & 1 \end{bmatrix} \begin{bmatrix} R_{3 \times 3} & \mathbf{t} \end{bmatrix} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = K[R|\mathbf{t}]\mathbf{X}_w$$

Generalizing intrinsics (1/2)

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f & & u_0 \\ & f & v_0 \\ & & 1 \end{bmatrix} [R_{3 \times 3} \quad \mathbf{t}] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = K[R|\mathbf{t}]\mathbf{X}_w$$

1. If pixels are not square?

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & & u_0 \\ & s_y & v_0 \\ & & 1 \end{bmatrix} [R_{3 \times 3} \quad \mathbf{t}] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = K[R|\mathbf{t}]\mathbf{X}_w$$

Method: High-Level Overview

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} s_x & & u_0 \\ & s_y & v_0 \\ & & 1 \end{bmatrix} [R_{3 \times 3} \quad \mathbf{t}]}_{\text{Camera projection matrix } M_{3 \times 4}} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = K[R|\mathbf{t}]\mathbf{X}_w$$

Camera projection matrix $M_{3 \times 4}$
(Note: we've also called this P before)

Method: High-Level Overview

$M = P = \text{projection matrix}$

$$\begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \approx \begin{bmatrix} m_{00} & m_{01} & m_{02} & m_{03} \\ m_{10} & m_{11} & m_{12} & m_{13} \\ m_{20} & m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}$$

$$u_i = \frac{m_{00}X_i + m_{01}Y_i + m_{02}Z_i + m_{03}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + m_{23}}$$

$$v_i = \frac{m_{10}X_i + m_{11}Y_i + m_{12}Z_i + m_{13}}{m_{20}X_i + m_{21}Y_i + m_{22}Z_i + m_{23}}$$

$$\begin{bmatrix} X_i & Y_i & Z_i & 1 & 0 & 0 & 0 & 0 & -u_iX_i & -u_iY_i & -u_iZ_i & -u_i \\ 0 & 0 & 0 & 0 & X_i & Y_i & Z_i & 1 & -v_iX_i & -v_iY_i & -v_iZ_i & -v_i \end{bmatrix} \begin{bmatrix} m_{00} \\ m_{01} \\ m_{02} \\ m_{03} \\ m_{10} \\ m_{11} \\ m_{12} \\ m_{13} \\ m_{20} \\ m_{21} \\ m_{22} \\ m_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Estimation of the full projection matrix is linear

but M has to be decomposed into $K, R, \mathbf{t}$ afterwards

(“QR decomposition” of first 3 columns of M to get K, R . Then solve for $\mathbf{t}$)

Generalizing intrinsics (2/2)

1. If pixels are not square?

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & & u_0 \\ & s_y & v_0 \\ & & 1 \end{bmatrix} [R_{3 \times 3} \quad \mathbf{t}] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} = K[R|\mathbf{t}]\mathbf{X}_w$$

2. If “radial distortions”, then the intrinsics can no longer be represented as a linear operator any more

Parametrizing radial distortion in large field-of-view cameras



Then, correct for radial distortion:

$$u_{\text{pre-distortion}}(r) = u_{\text{post-distortion}}(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 + \dots)$$

$$v_{\text{pre-distortion}}(r) = v_{\text{post-distortion}}(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 + \dots)$$

where r is the distance from a (usually unknown) image center location (u_0, v_0) .

Can choose the degree of the radial distortion to calibrate for. More => more accurate, but requires more images to fit well.

What Does (Intrinsics) “Calibration” Do?

Estimates the *intrinsic parameters*

- f focal length (or s_x, s_y to allow non-square pixels)
- (u_o, v_o) Principle point
- $k_1, k_2, \dots$ radial distortion parameters

Colab worksheet (Camera Calibration with OpenCV)!

- <https://colab.research.google.com/drive/1Vf6pBFo84OKR0E3j3t3HJjEN71o6k0sk?authuser=0>
- (Optional) Also see OpenCV Tutorial on Calibration.

Camera Calibration Flowchart

Define real world coordinates of 3D points using checkerboard pattern of known size.



Capture the images of the checkerboard from different viewpoints.



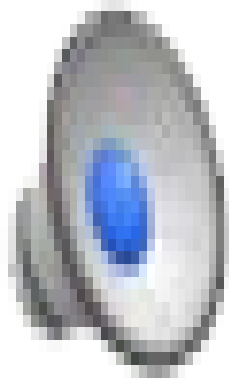
Use **findChessboardCorners** method in OpenCV to find the pixel coordinates (u, v) for each 3D point in different images



Find camera parameters using **calibrateCamera** method in OpenCV, the 3D points, and the pixel coordinates.

<https://learnopencv.com/camera-calibration-using-opencv/>

Output: Undistorted images and video!



Summary of Tools for Single-View Geometry (1/2)

- Finding vanishing points:
 - Using parallel lines
 - Using homographies
 - Using “cross ratios”
- Using VPs to compute horizon and determine camera orientation w.r.t. ground (assuming standard intrinsics K)
- Finding homographies from planes:
 - from 2 orthogonal VPs plus correspondences for the 2 points $(0, 0)$, and $(1, 1)$.
 - from 4 non-square points using inversion and composition of 2 homographies.
 - from ≥ 4 arbitrary non-collinear point correspondences.
- Using homographies for:
 - Virtual billboard display
 - Finding camera pose and refining with Kabsch (given intrinsics K)

Summary of Tools for Single-View Geometry (2/2)

- Using cross ratios to measure real lengths:
 - Along a line, given 2 world lengths.
 - Along a line, given 1 world length and the VP.
 - Using VPs on ground plane, vertical VP, and vertical cross ratios for length transfer.
- Using pose and intrinsics (full projection matrix) to project AR objects onto world.
- Finding extrinsics from 3 2D->3D point correspondences (P3P problem)
 - What if we had depth images?
 - What if we only had standard RGB images?
- Finding extrinsics from $n > 3$ 2D->3D point correspondences not on a plane (PnP problem)
- Finding camera intrinsics (camera calibration):
 - Finding intrinsics K from 3 orthogonal VPs (camera calibration part 1)
 - Overview of more general camera intrinsics calibration under radial distortion

Next, onwards to 2 Views!